SPE-226887-MS

Improved Methods for Determining Maximum Horizontal Stress from Borehole Breakout

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

In-situ stresses are among the most important inputs in all deep underground engineering analysis. In the oil and gas industry, in-situ stresses are critical to the integrity of boreholes during drilling, completion, and production. Uncertainty in stress determination can result in costly failures, such as borehole collapse, sand production, liner deformation, etc., especially when the safe operation window is narrow, like in underbalanced drilling. In this work, we validated two improved methods for accurately determining the maximum horizontal stress from borehole breakout. Compared with the original breakout method, the major difference is the net borehole pressure term, i.e., in an impermeable borehole, the effect of net borehole pressure is magnified by friction angle thus the original breakout method tends to underestimate the maximum horizontal stress; in a permeable borehole, the effect of net borehole pressure is scaled by Poisson's ratio of the formation. The predicted maximum horizontal stress is further constrained by the limit imposed by the stress regime where the borehole is located. The improved methods are verified with a numerical simulation and shown to be capable of improving the accuracy of the determined maximum horizontal stress from breakout width. Validation with a field case is also conducted. The improved methods are ready for field deployment, e.g., to support the design of some new operation projects, or diagnose observations and failures of some existing operation projects.

Introduction

Accurate information on in-situ stresses is critical to the reliable design and analysis for all deep underground engineering projects. In the oil and gas industry, in-situ stresses are critical to the integrity of boreholes during drilling, completion, and production. Uncertainty in stress determination can result in disastrous failures, such as borehole collapse, catastrophic sand production, and large liner deformation.

There are three principal in-situ stresses that need to be determined, i.e., vertical stress, maximum horizontal stress, and minimum horizontal stress. Unlike the vertical stress, which may be obtained by integrating densities of geological layers (Tingay et al. 2003), and minimum horizontal stress, which may be measured by in-situ tests like MiniFrac (Woodland et al. 1989), the maximum horizontal stress cannot be directly measured, instead it must be inferred from other measurements. One approach is to inversely calculate the maximum horizontal stress from borehole breakout angle measured by caliper log or borehole televiewer (Zoback et al. 1985; Barton et al. 1988; Zoback 2010). The feasibility of the approach was confirmed by the laboratory experiments (Haimson and Herrick 1986). Due to its clear concept and simplicity in application, this method has then been extensively adopted in the determination of the maximum horizontal stress in the last four decades (Vernik et al. 1992; Amadei et al. 1997; Shen 2008; Reis et al. 2013; Zhang et al. 2017; Baouche et al. 2024).

In an attempt to evaluate how the permeability condition of the borehole affects the applicability of this borehole breakout model proposed in Barton and Zoback (1988), it was found that the model can be further refined for both impermeable and permeable boreholes. In the next section, we will present two improved models, one for impermeable boreholes and the other for permeable boreholes, that are expected to determine the maximum horizontal stress from borehole breakout more accurately. The predicted maximum horizontal stress is further constrained by the limit imposed by the stress regime where the borehole is located. In section 3, numerical simulation is performed to demonstrate the improved models in improving the accuracy of the determined maximum horizontal stress from the breakout width. In section 4, the improved models are applied and validated in a field case study. Two improved models have been implemented in an in-house wellbore mechanics software and ready for testing practical cases. In Section 5, a concluding summary is given to close this paper.

Borehole Breakout Models

Existing model

Fig. 1 illustrates the breakouts around a borehole induced during the drilling process. Breakouts are essentially compressively shearing yielding regions. They are primarily determined by the interplay of the in-situ stresses, borehole conditions, and formation properties. For example, if the in-situ stress is low enough, the borehole pressure is high enough, or the formation is strong enough, a breakout will not occur.

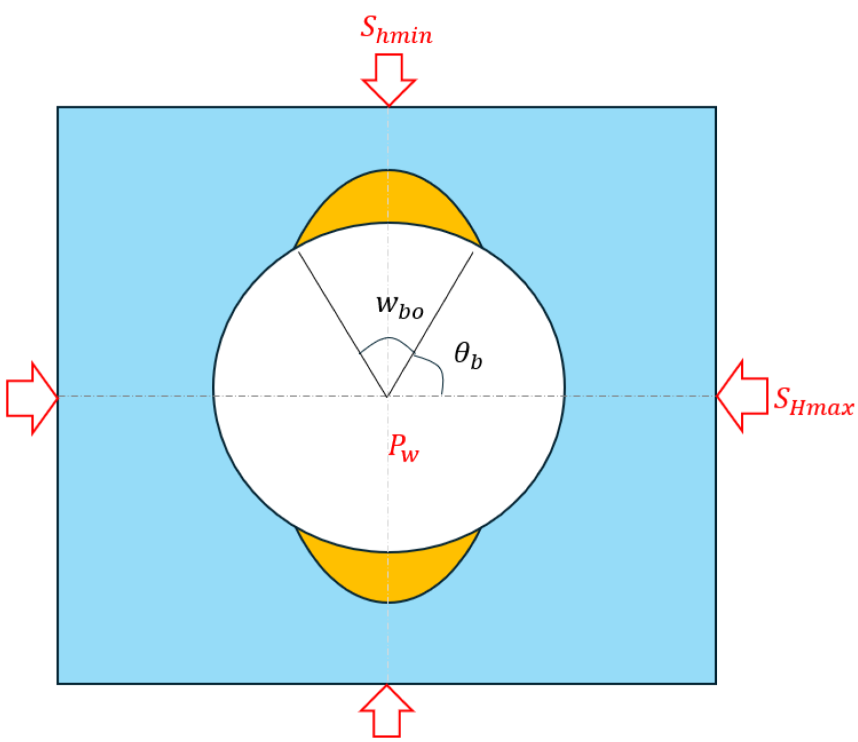

Figure 1—Borehole breakouts induced during drilling process.

In a case where a breakout does occur, the following equation is perhaps the most widely used model to estimate the maximum horizontal stress from borehole breakout (Barton et al. 1988; Zoback 2010):

$$S_{Hmax} = \frac{UCS + \Delta P_W + 2P_p}{1 - 2\cos 2\theta_b} - S_{hmin}\frac{1 + 2\cos 2\theta_b}{1 - 2\cos 2\theta_b} \tag{1}$$

where S_{Hmax} is the maximum horizontal stress; S_{hmin} is the minimum horizontal stress; UCS is the uniaxial compressive strength of the formation rock; ΔP_W is the net fluid pressure inside the borehole, i.e., $\Delta P_W = P_W - P_p$, where P_W is the borehole pressure and P_p is the pore pressure in the far field; θ is the borehole angle measured from the maximum horizontal stress; θ_b is the borehole angle where the rock mass on the borehole wall starts to run into plastic yielding; w_{bo} is the breakout width (angle), i.e., $w_{bo} = \pi - 2\theta_b$.

Improved models

Attempting to derive the model equation in Eq. (1) using PoroMechanics theory (see Appendix A), we found that some important mechanisms were missing in it. Accordingly, two refined models were proposed, one for impermeable boreholes, and the other for perfectly permeable boreholes.

a. Impermeable boreholes

For impermeable boreholes, derivation indicates that the friction angle is a relevant parameter in the maximum horizontal stress determination:

$$S_{Hmax} = \frac{UCS + \frac{2}{1-\sin\varphi}\Delta P_W + 2P_p}{1 - 2\cos 2\theta_b} - S_{hmin}\frac{1 + 2\cos 2\theta_b}{1 - 2\cos 2\theta_b} \tag{2}$$

Note, if the net borehole pressure (ΔP_W) in Eq. (2) is equal to zero (borehole is at the balanced condition), it will be equivalent to Eq. (1). If the net borehole pressure (ΔP_W) is greater than zero (i.e., the borehole is in overbalanced condition, which is commonly encountered in the borehole drilling), the effect of net borehole pressure (ΔP_W) will be underestimated if Eq. (1) is used.

b. Perfectly permeable boreholes

For a permeable borehole in a perfectly permeable formation, the borehole fluid will infiltrate into the formation and the pore pressure in the formation around the borehole will increase towards the level of borehole fluid pressure. In this case, the breakout model in Eq. (1) needs to be refined as:

$$S_{Hmax} = \frac{UCS + \frac{2v}{1-v}\Delta P_W + 2P_p}{1 - 2\cos 2\theta_b} - S_{hmin}\frac{1 + 2\cos 2\theta_b}{1 - 2\cos 2\theta_b} \tag{3}$$

Unlike Eq. (1), which shows the irrelevance of Poisson's ratio in determining the maximum horizontal stress from breakout width, Eq. (3) shows that Poisson's ratio will affect the calculation of the maximum horizontal stress from the breakout width in a permeable borehole. Interestingly, Eq. (3) will reduce to the original breakout equation, Eq. (1), in a formation where Poisson's ratio is equal to 1/3.

It should be noted that, for the exposure time of interest, if the fluid flow field around a borehole is in a transient state, the pore pressure and subsequently the effective stress in the near-borehole region will be time-dependent, Eq. (3) will be inapplicable, and a coupled hydromechanical analysis will be necessary.

Constraints of stress regime

In normal stress regimes, where $S_V \geq S_{Hmax} \geq S_{hmin}$, S_V and S_{hmin} shall satisfy the following relationship (Zoback 2010):

$$\frac{\sigma_1}{\sigma_3} = \frac{S_V - P_p}{S_{hmin} - P_p} \leq \left[\left(\mu^2 + 1\right)^{1/2} + \mu\right]^2 \tag{4}$$

where σ_1 and σ_3 are the maximum and minimum effective principal stress, respectively; μ is the frictional coefficient, $\mu = \tan\varphi$; φ is the friction angle. It should be noted that Eq. (4) is essentially a special case of Mohr-Coulomb failure criterion. That is, the standard Mohr-Coulomb failure criterion may be expressed as:

$$\sigma_1 \le \sigma_3 \frac{1+\sin\varphi}{1-\sin\varphi} + \frac{2c\cos\varphi}{1-\sin\varphi} \tag{5}$$

where c is the cohesive strength. Eq. (5) will reduce to Eq. (4) with c set to zero (arguably this assumption is justifiable because the cohesive strength should be negligibly low during the geological sedimental process for most rock layers) and some simple mathematical manipulation, that is:

$$\sigma_1 \le \sigma_3 \frac{1+\sin\varphi}{1-\sin\varphi} = \sigma_3 \frac{(1+\sin\varphi)^2}{\cos^2\varphi} = \sigma_3 \left[\frac{1}{\cos\varphi} + \tan\varphi\right]^2 = \left[\left(\mu^2+1\right)^{1/2} + \mu\right]^2 \tag{6}$$

Similarly, in strike-slip stress regimes, where $S_{Hmax} \ge S_V \ge S_{hmin}$, S_{Hmax} and S_{hmin} shall satisfy the following relationship:

$$\frac{\sigma_1}{\sigma_3} = \frac{S_{Hmax} - P_p}{S_{hmin} - P_p} \le \left[\left(\mu^2+1\right)^{1/2} + \mu\right]^2 \tag{7}$$

And, in reverse stress regimes, where $S_{Hmax} \ge S_{hmin} \ge S_V$, S_{Hmax} and S_V shall satisfy the following relationship:

$$\frac{\sigma_1}{\sigma_3} = \frac{S_{Hmax} - P_p}{S_V - P_p} \le \left[\left(\mu^2+1\right)^{1/2} + \mu\right]^2 \tag{8}$$

The maximum horizontal stress (S_{Hmax}) determined from breakout models needs to be checked against these constraints imposed by the stress regime to ensure they are not violated.

Numerical Verification

In this section, we will perform a numerical simulation to model the borehole drilling process, measure the breakout from the simulation result, back-calculate the maximum horizontal stress from the breakout width, and compare the predictions of the existing breakout model and the improved breakout models. The relevance of the breakout depth to the maximum horizontal stress determination will also be experimented and discussed.

The numerical simulation will be carried out in FLAC, a commercial computational geomechanics software designed for simulating the behavior of structures built of soil, rock, and other materials that may undergo plastic flow when their yield limits are reached. When modeling a problem in FLAC, material regions are discretized into elements (zones); all elements are connected and adjusted to fit the object to be modeled. Each element responds to the boundary restraints, such as applied forces, velocities, and accelerations, following the linear or nonlinear constitutive law of the material assigned to it. The material can yield and flow, and the grid can deform and move with the material (Itasca 2011).

In FLAC, the elastoplastic mechanical analysis can be conducted in static or dynamic analysis mode, in small-strain or large-strain mode. The fluid flow calculation allows both transient and steady-state analysis. Mechanical and fluid flow calculations can be performed separately, or in a loosely coupled manner, or in a fully coupled manner.

One salient feature of FLAC is that it provides a built-in programming language, FISH, which allows users to fully control nearly every aspect of the simulation, ranging from mesh generation and boundary condition enforcement to computation process interference and user-defined constitutive model development. In this study, FISH is used to generate mesh, control the pressure reduction inside the borehole, realize the progressive removal of rock mass from the breakout region, and post-process the simulation results.

Input data

In this verification example, we are simulating the process of drilling a vertical borehole. The in-situ stresses and pore pressure, the formation's mechanical properties and borehole condition that are used in the numerical model are provided in Table 1.

Table 1—In-situ condition, formation properties and well condition

In-situ condition	Formation properties	Well condition
S_V= 45 MPa	E = 41.5 MPa	r_w = 10 cm
S_{Hmax}= 40 MPa	v = 0.27	ΔP_W = 2 MPa
S_{hmin} = 30 MPa	c = 15 MPa	
P_p= 10 MPa	φ = 30 deg.	

Since the stiffness of the rock matrix is much less than that of rock grains after rock runs into plastic yielding, Biot's coefficient of effective stress (α) may be assumed to be equal to 1 in failure analysis. In this case, the effective stresses and UCS of the formation can be evaluated as:

$$\sigma_V = S_V - \alpha P_p = 35\,\text{MPa}$$
$$\sigma_{Hmax} = S_{Hmax} - \alpha P_p = 30\,\text{MPa}$$
$$\sigma_{hmin} = S_{hmin} - \alpha P_p = 20\,\text{MPa}$$
$$UCS = \frac{2c\cos\varphi}{1-\sin\varphi} = 52\,\text{MPa}$$

Simple mathematical work confirms that the relationship between vertical stress and minimum horizontal stress does not violate the constraint of Eq. (4) imposed by the normal stress regime.

Mesh generation

Considering the symmetric nature of the problem, a quarter-donut mesh was sufficient for modeling this problem. Here we chose to use a half-donut mesh, expecting to observe a symmetric response for double-checking the correctness in the model setup and the system's performance.

The computational mesh is generated in two steps. As illustrated in Fig. 2, in the first step a rectangular mesh is generated. In the second step, the grid-point coordinates of elements are mapped to the borehole geometry and surrounding formation extensions as desired, which can be painlessly accomplished using a FISH function.

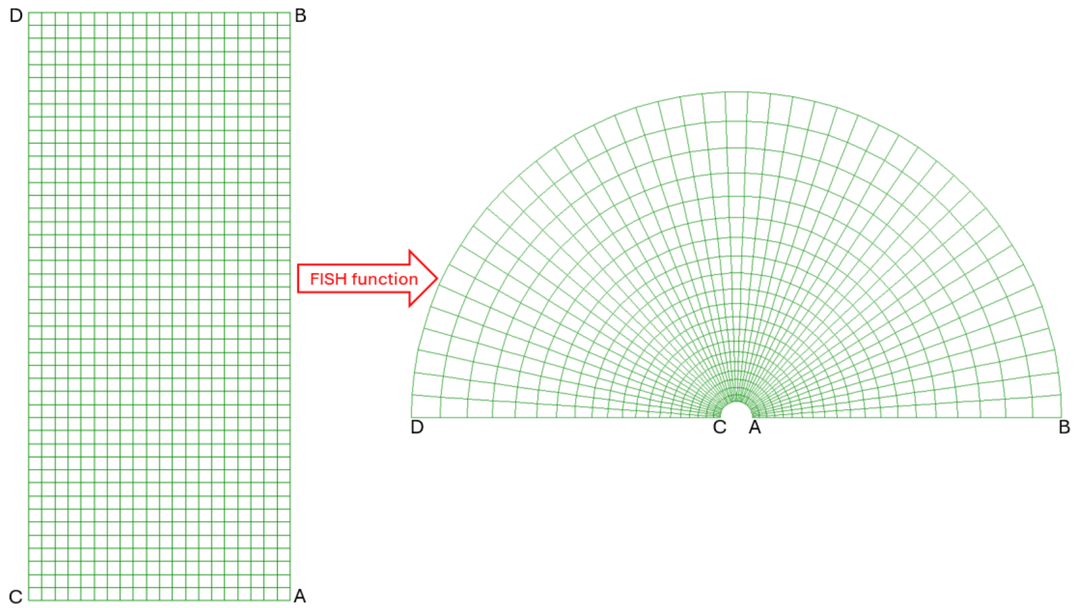

Figure 2—Procedure of generating computational mesh for modeling borehole problems.

In this particular model, we assume that the borehole radius is 10 cm, the extension of the rock mass around the borehole (representing far-field) is 2 m away from the borehole center. An 80 × 180 mesh is employed in this simulation. In the radial direction, the sizes of 80 elements are increasing at a constant ratio of 1.08. In the circumferential direction, 180 elements are uniformly distributed. The computational mesh that is used in the simulation is displayed in Fig. 3.

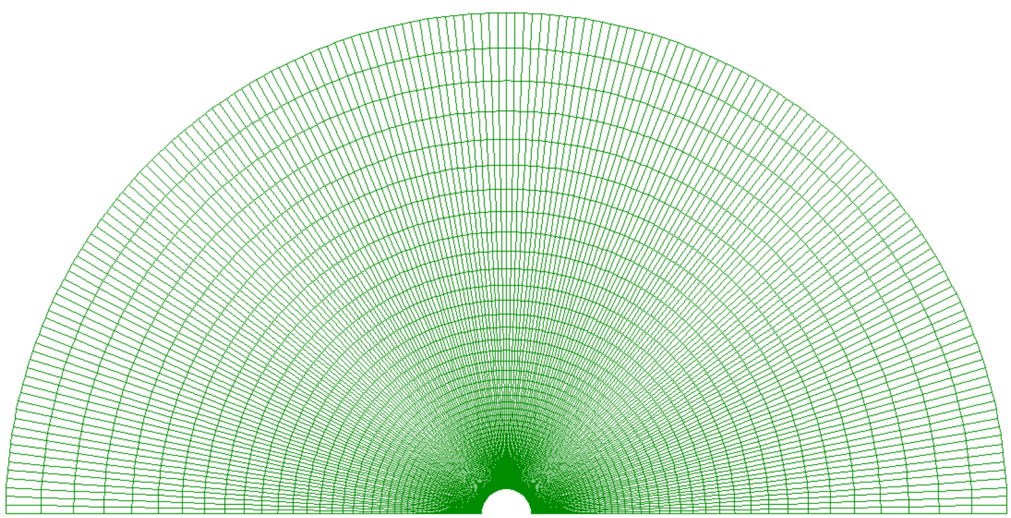

Figure 3—Computational mesh used in the simulation.

Simulation procedure

After the mesh is generated, the Mohr-Coulomb type material with mechanical properties tabulated in Table 1 is assigned to all the elements in the mesh. Next, the stress components SXX, SYY, and SZZ are initialized to 35 MPa, 30 MPa, and 25 MPa, respectively. Note that, here we are modeling an impermeable borehole, in which there is no fluid flow taking place around the borehole; to simplify the model setup and execution, the problem is modeled in effective stress space.

On the symmetric lines, $\overline{AB}$ and $\overline{CD}$, the roller boundary is enforced. At the outer boundary, normal stresses of 30 MPa and 25 MPa are applied in x- and y- directions, respectively, as far-field stresses (e.g.,

$\widehat{DB}$ in Fig. 2). The borehole drilling, which may be assumed to be a pseudo-static process, is simulated by progressively reducing the pressure acting on the borehole surface (e.g., $\widehat{CA}$ in Fig. 2) from the initial in-situ stress to the drilling mud pressure. This is achieved by enforcing the borehole pressure to linearly decrease over a large number of calculation steps:

$$P_w = \begin{cases} P_{ini} - \left(P_{ini} - P_{mud}\right)\frac{step-step0}{nstep}, & step < step0 + nstep \\ P_{mud}, & step \geq step0 + nstep \end{cases} \tag{9}$$

where P_w is the pressure applied on the borehole surface; P_{ini} is the pressure imposed on the borehole surface when the drilling process starts it is set to the mean effective stress at the initial state; P_{mud} is the net drilling mud pressure, which was set to 2 MPa in this model; *step0* is the calculation step when the drilling process starts; *nstep* is a large number representing the time lapse taken to crush and completely loosen the rock mass in the drilling process, it is set to 10,000 in this simulation; *step* is the current calculation step. This function can be implemented in the numerical model via a FISH function in a straightforward way.

Simulation results

The accuracy in predicting plastic yielding and breakout region depends on the accuracy of the model in calculating the stress around the borehole. The accuracy of stress solution can be affected by many factors, such as mesh resolution, boundary extension, modeling approach, human errors, etc.

Stress solution. For a cylindrical hole sitting in a hydrostatic stress field, the elastoplastic solutions of stress and deformation around the borehole can be analytically evaluated, e.g., using the solution proposed in Salencon (1969). When the in-situ stress condition becomes anisotropic, i.e., the two confining principal stresses are unequal, no analytical solution exists for calculating elastoplastic stress or displacement.

Since the analytical solution for stress around a borehole can be evaluated using the Kirsch solution if the rock mass around the borehole is an elastic medium, we can set the cohesive strength of the Mohr-Coulomb type material to a very high value to prevent plastic yielding. This way the material will behave like an elastic medium because stress will never reach the yielding surface.

Fig. 4 compares the tangential stress distribution around the borehole calculated by the numerical model with its analytical counterpart given by Kirch solution at the net borehole pressure (ΔP_W) of 2 MPa:

$$\sigma_{\theta\theta} = \left(\sigma_{Hmax} + \sigma_{hmin}\right) - 2\left(\sigma_{Hmax} - \sigma_{hmin}\right)\cos 2\theta - \Delta P_W \tag{10}$$

where θ starts counterclockwise from the X-axis, i.e., the line $\overline{AB}$ in Fig. 2. As seen, the numerical result and analytical solution agree well at all points (centers of all elements). It should be noted that in FLAC, the sign convention for stress is positive in tension and negative in compression. Calculation indicates that the maximum error between them is 0.21%, and the average error is 0.02%. This verifies, partly at least, that the adopted mesh has sufficient resolution, and the numerical model has been set up and executed appropriately.

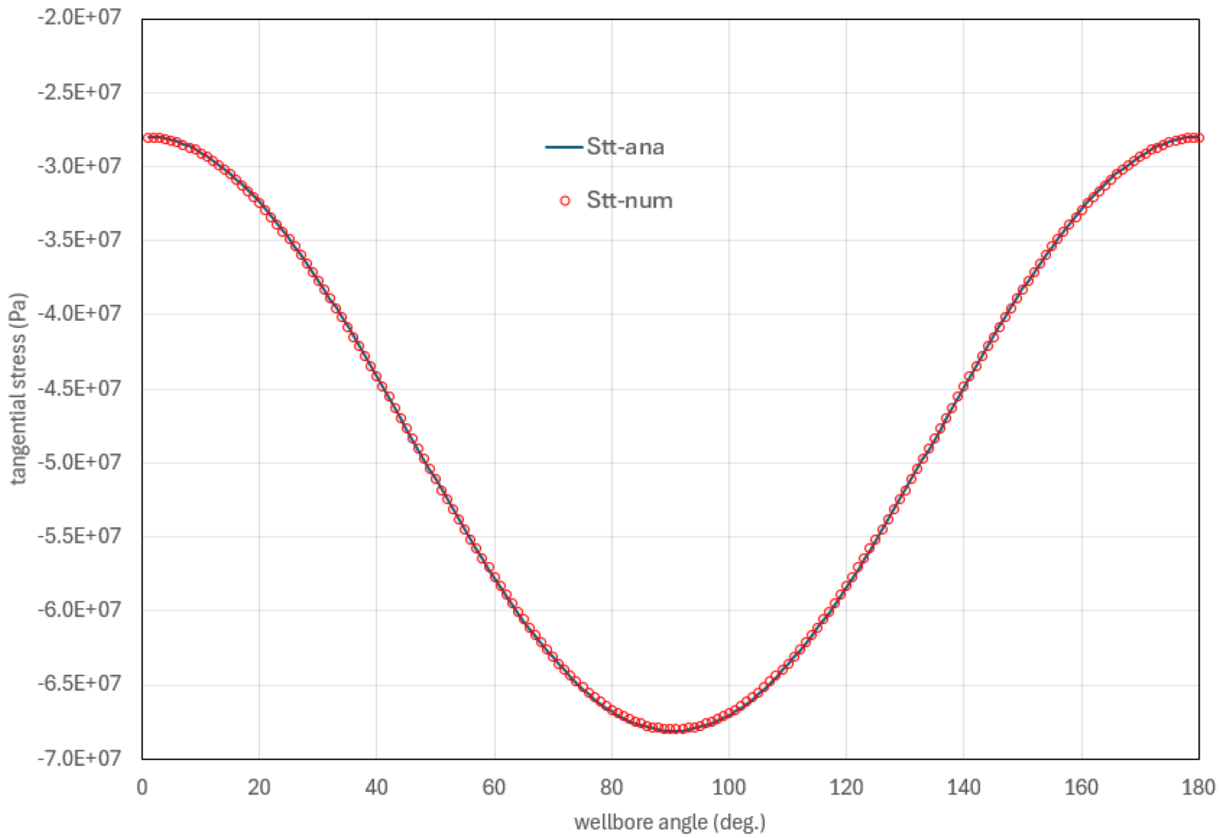

Figure 4—Distribution of tangential stress around borehole (comparison between analytical solution and simulation result).

Fig. 5 displays the contours of radial stress and tangential stress around the borehole at net borehole pressure (ΔP_W) of MPa. Unlike the chart plot in Fig. 4, which gives numeric values on the borehole surface, Fig. 5 gives the distributions of calculated stress at all elements in the model. As can be seen, because the maximum horizontal stress is in x-direction, the highest concentration of tangential stress occurs in the crown region, where the breakout will initiate when the borehole pressure is low enough such that the tangential stress exceeds the compressive strength that is decided by the cohsive strength, friction angle and radial stress (i.e., the minimum principal stress). The radial stress contour verifies that the radial stress on the borehole surface is equivalent to the borehole pressure of 2 MPa.

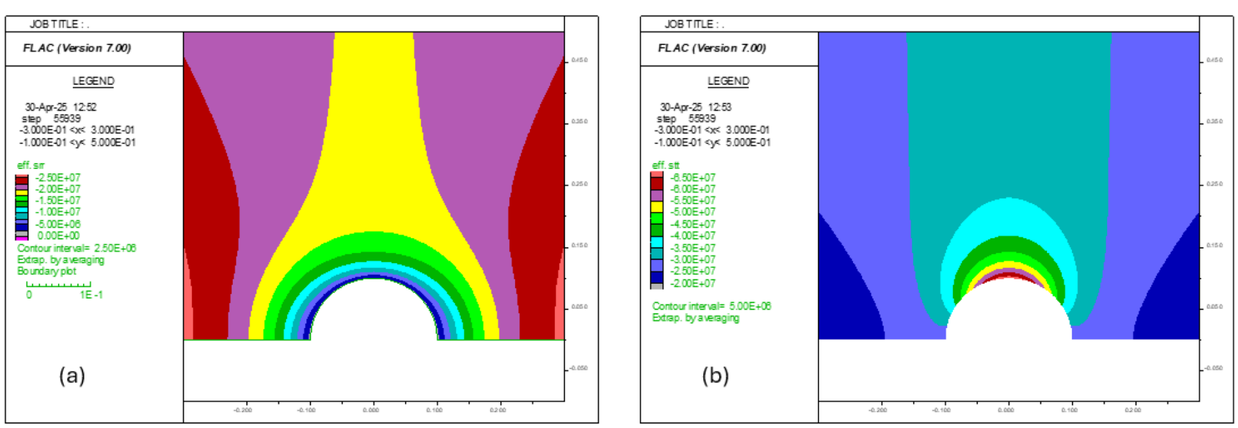

Figure 5—Contours of (a) radial stress and (b) tangential stress at net borehole pressure of 2 MPa (elastic analysis).

Breakout analysis. The model is re-run with cohesive strength reset to its real value of 15 MPa. The contours of radial stress and tangential stress resulted from the elastoplastic analysis are provided in Fig. 6. Compared with the elastic analysis in Fig. 5, the tangential stress shows an obvious difference; thus, plastic deformation must have taken place at a net borehole pressure of 2 MPa.

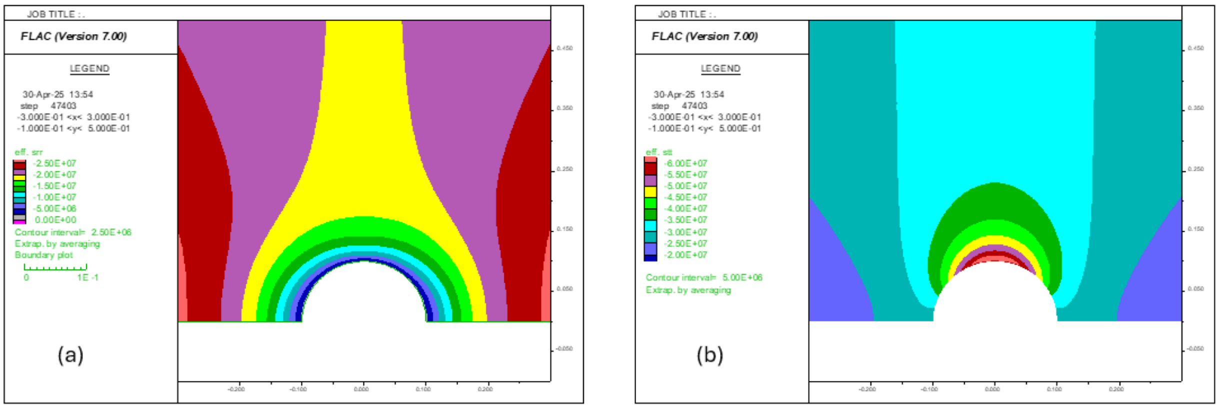

Figure 6—Contours of (a) radial stress and (b) tangential stress at net borehole pressure of 2 MPa (elastoplastic analysis).

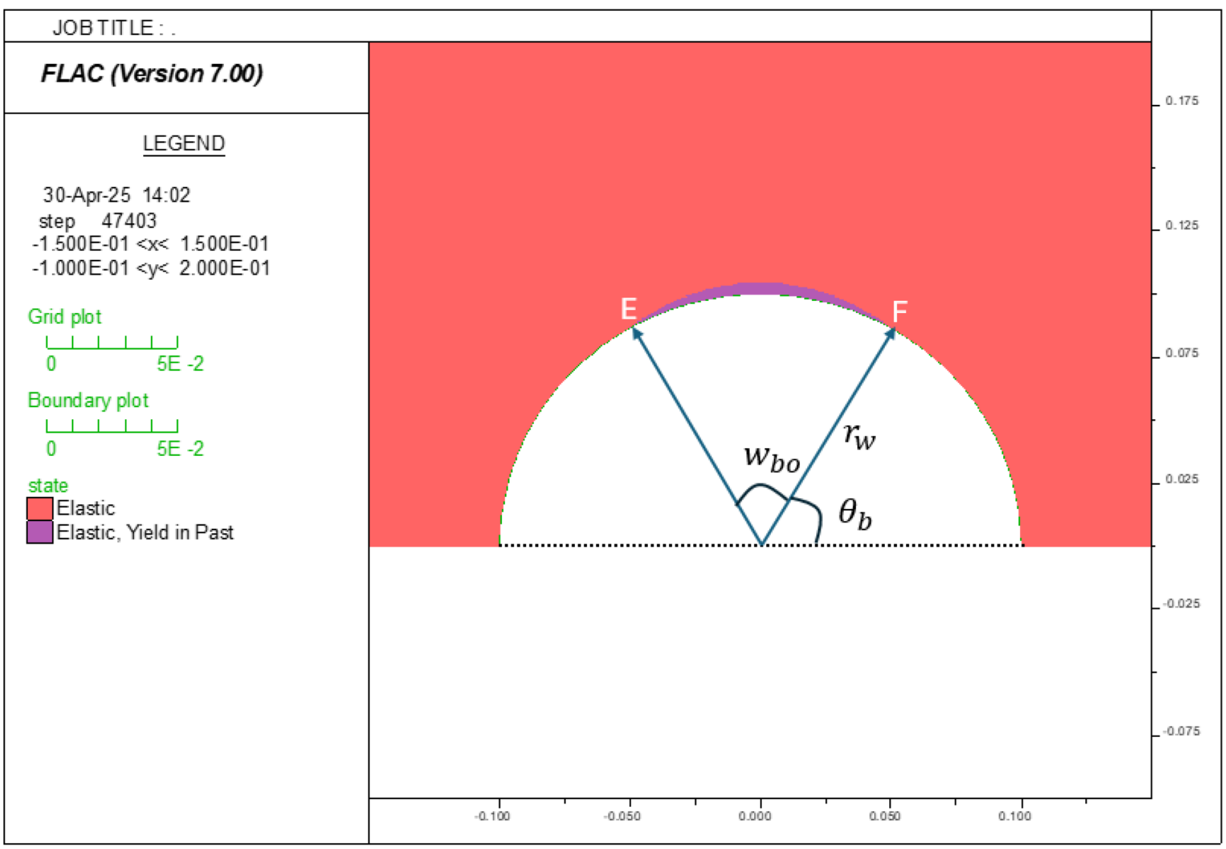

Figure 7—Plastic yielding region around borehole at net borehole pressure of 2 MPa (elastoplastic analysis).

The plastic yielding region at a net borehole pressure of 2 MPa is presented in Fig. 6. The breakout width, defined by two tips of the breakout region, can be measured from this plot. Alternatively, a more precise way to measure it is to write a FISH function to scan each element on the borehole surface to identify the coordinates of two elements (i.e., 'E' and 'F' in Fig. 6) at the two tips. The breakout angle can be calculated from the distance ($\overline{EF}$) and borehole radius (r_w):

$$w_{bo} = 2\sin^{-1}\frac{\overline{EF}}{2r_w} \tag{11}$$

Scanning and calculation of the FISH function show that the breakout angle (w_{bo}) is 59°; accordingly, $2\theta_b = 121°$.

Now we pretend that the maximum horizontal stress is not known, and we need to estimate S_{Hmax} from the breakout angle. Substituting other known parameters in Table 1 into Eq. (1) shows that S_{Hmax} = 36.9 MPa. On the other hand, Eq. (2) shows that 39.9 MPa, which is much closer to the "correct answer" of 40 MPa.

Effect of breakout depth. After borehole breakouts form, their depths are expected to increase, but their width shall stay unchanged, as confirmed theoretically in Zoback (2010) and experimentally in Haimson and Herrick (1989). In the numerical model in FLAC, we can simulate the evolution of borehole breakouts with a bit of additional work: (1) remove the boundary condition on the borehole surface (i.e., arc surface $\widehat{AC}$; (2) remove all the elements that have run into plastic yielding; (3) apply the borehole pressure to new arc surface $\widehat{AC}$; (4) run the model to mechanical equilibrium again. This process is repeated until no single element in the whole model runs into plastic yielding. At the end of the simulation, the breakout shape predicted by FLAC numerical model is displayed in Fig. 8-(b), which looks similar to the observation in the lab experiment reported in Haimson and Herrick (1989).

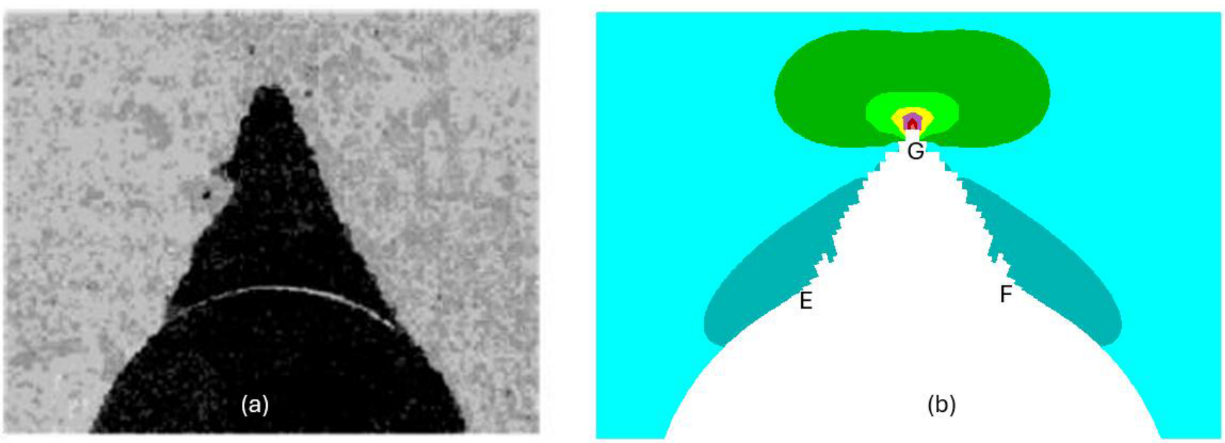

Figure 8—Evolution of breakout observed in (a) lab experiment (Haimson and Herrick 1989) and (b) FLAC numerical model (tangential stress contour also shown).

It should be emphasized that this idealized model simply demonstrates that the breakout depth could be decoupled from the breakout width in an extreme case. In reality, most probably the rock mass inside the plastic region would not completely fall off because of residual strengths, i.e., for many rocks, the cohesive strength may decrease, but the frictional resistance would increase as the plastic deformation progresses. With these post-yield softening and hardening mechanical behaviors considered, the breakout depth may be predicted from the in-situ stresses (Walton et al. 2015).

Perfectly permeable borehole. If the time that formation is exposed to drilling pressure is significantly larger than the borehole's fluid diffusion characteristic time, the borehole may be assumed to be perfectly permeable. In this case, the confining stresses around borehole will be changed by injection/depletion effect. In this process the variations in effective stresses will be:

$$\Delta\sigma_V = -\alpha\Delta P_w \quad \Delta\sigma_{Hmax} = \Delta\sigma_{hmin} = \alpha\frac{\nu}{1-\nu}\Delta P_w \tag{12}$$

To model this problem using the model presented above, the following changes need to be made: (1) initialize the stress state using the adjusted in-situ stresses; (2) apply the adjusted stresses (i.e., σ_V = 33

MPa; σ_{Hmax} =29.3 MPa; σ_{hmin} = 19.3 MPa) on the outer boundary; (3) set the target mud pressure inside the borehole to zero (i.e., P_w = 0 MPa).

Fig. 9 presents the contours of radial stress and tangential stress around the borehole given by the simulation for the case of a perfectly permeable borehole. Comparison with Fig. 6 shows that both the minimum radial stress (on the borehole surface) and maximum tangential stress (in the crown region) in the case of a permeable borehole are lower than the corresponding values in the case of an impermeable borehole. This observation implies two mechanisms, i.e., on one hand, decreasing maximum tangential stress will alleviate the breakout, on the other hand, decreasing radial stress on the borehole surface will reduce the confining stress, thus intensifying the breakout. The competition between these two effects will decide the eventual result.

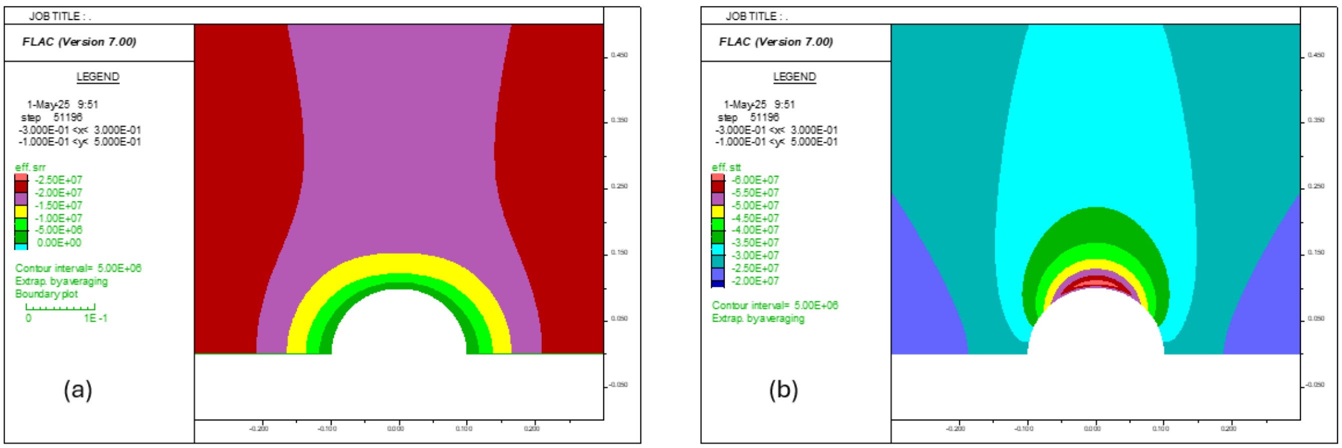

Figure 9—Contours of (a) radial stress and (b) tangential stress at net borehole pressure of 2 MPa (permeable borehole).

Measuring the plastic zone on the simulation result reports that the breakout width (w_{bo}) is 80.1°; accordingly, $2\theta_b$ = 99.9°. Evidently, the effect of decreasing confining stress is the dominant mechanism, so the breakouts increase because of the high permeability of the borehole. Pretending that the maximum horizontal stress is not known, we wish to estimate S_{Hmax} from breakout width. Substituting other known parameters in Table 1 into Eq. (1) shows that S_{Hmax} = 40.4 MPa. On the other hand, Eq. (3) predicts that S_{Hmax} = 40 MPa, which is identical to the "correct answer" of 40 MPa.

Validation With Field Data

In this section, the improved models are applied to determine the maximum horizontal stress from a deep well in a real case. After drilling, the breakouts at different depths inside the well were inferred from the caliper log. It was approximated that the average breakout angle is between 50° and 60°. The in-situ condition (minimum horizontal stress and pore pressure), formation properties (UCS, friction angle, Poisson's ratio), and well conditions (borehole pressure P_w) at the four different depths are provided in Table 2.

Note that, as shown in Table 2, the borehole pressure was lower than the formation pressure during the drilling, so this well was drilled under an underbalanced drilling condition. Assuming a breakout width of 50.0° for the four depths, Table 3 shows the results of Eq. (1), Eq. (2), and Eq. (3) along with the reference (actual) S_{Hmax} values. Clearly, Eq. 3 (permeable borehole model) is the closest to the reference values.

Table 2—In-situ conditions and formation properties at four depths in Well ABC

TVD (ft.)	In-situ condition	Formation properties	Well condition
Depth (1)	S_{hmin}= 12715 P_p= 7994	UCS = 20610 v = 0.275 φ = 33.6^o	P_w = 7314.47
Depth (2)	S_{hmin}= 12408 P_p= 8003	UCS = 18256 v = 0.298 φ = 35.4°	P_w = 7314.47
Depth (3)	S_{hmin}= 12961 P_p= 8096	UCS = 20004 v = 0.287 φ = 38.9°	P_w = 7314.47
Depth (4)	S_{hmin}=13034 P_p= 8131	UCS = 20610 v = 0.285 φ = 42.6°	P_w = 7314.47

Table 3—Determination of S_{Hmax} with fixed breakout width (w_{bo} = 50.0°)

Depth	Eq. (1) (Barton & Zoback)	Eq. (2) (Impermeable borehole)	Eq. 3 (Permeable Borehole)	Reference (actual) value
Depth (1)	17347	16463	17407	17874
Depth (2)	16286	15332	16324	16434
Depth (3)	17199	16069	17249	17783
Depth (4)	17502	16159	17554	18169

In another attempt, we tried to calibrate the breakout widths against the reference S_{Hmax} values for the four depths. This exercise yielded the following breakout angles for the four different depths: w_{bo} = 57.3° (Depth 1), w_{bo} = 52.3° (Depth 2), w_{bo} = 58.9° (Depth 3), w_{bo} = 59.6° (Depth 4). Substituting the parameters in Table 2 along with the calibrated breakout widths into Eq. (1), Eq. (2), and Eq. (3), the results are shown in Table 4. Evidently, the improved model for permeable borehole can give a very reliable estimate for S_{Hmax} for the four different depths.

Table 4—Determination of S_{Hmax} with calibrated breakout width (w_{bo})

Depth	Eq. (1) (Barton & Zoback)	Eq. (2) (Impermeable borehole)	Eq. 3 (Permeable Borehole)	Reference (actual) value
Depth (1)	17806	16835	17873	17874
Depth (2)	16397	15415	16436	16434
Depth (3)	17728	16456	17784	17783
Depth (4)	18109	16583	18169	18169

Concluding summary

The existing model for computing maximum horizontal stress from borehole breakout cannot accurately compute the maximum horizontal stress from borehole breakout width because it overlooks some primary affecting factors. Its drawback is addressed in two improved models that were derived based on PoroMechanics theory.

In this work, the improved models were verified through comparison with their numerical counterparts constructed in the commercial geomechanics software FLAC. Numerical simulation shows that the maximum horizontal stress can be reliably estimated from the borehole breakout width for both impermeable boreholes and perfectly permeable boreholes in rocks that obey Mohr-Coulomb failure criterion; the breakout depth could be irrelevant to the maximum horizontal stress. The determined maximum horizontal stress also needs to respect the constraint of the stress regime where the borehole is located.

Furthermore, the derived improved models were also validated against a real underbalanced drilled well. Four depths were selected, and the corresponding breakout angles were determined from caliper logs. The obtained results show that our improved model for permeable boreholes outperforms the traditional borehole breakout model (Barton et al. 1988; Zoback 2010). The traditional borehole breakout model tends to underestimate the maximum horizontal stress in impermeable boreholes and could overestimate or underestimate the maximum horizontal stress in permeable boreholes, depending on the Poisson's ratio of the formation.

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Appendix A
Solution Derivation

In effective stress space, the loading condition for an impermeable borehole is illustrated in Fig. A1, in which σ_H and σ_h are the maximum and minimum horizontal effective stress, respectively; P_p is the original formation pore pressure; ΔP_w is the net wellbore pressure; w_{bo} is the breakout width (angle); θ_b is the wellbore angle where the breakout region starts.

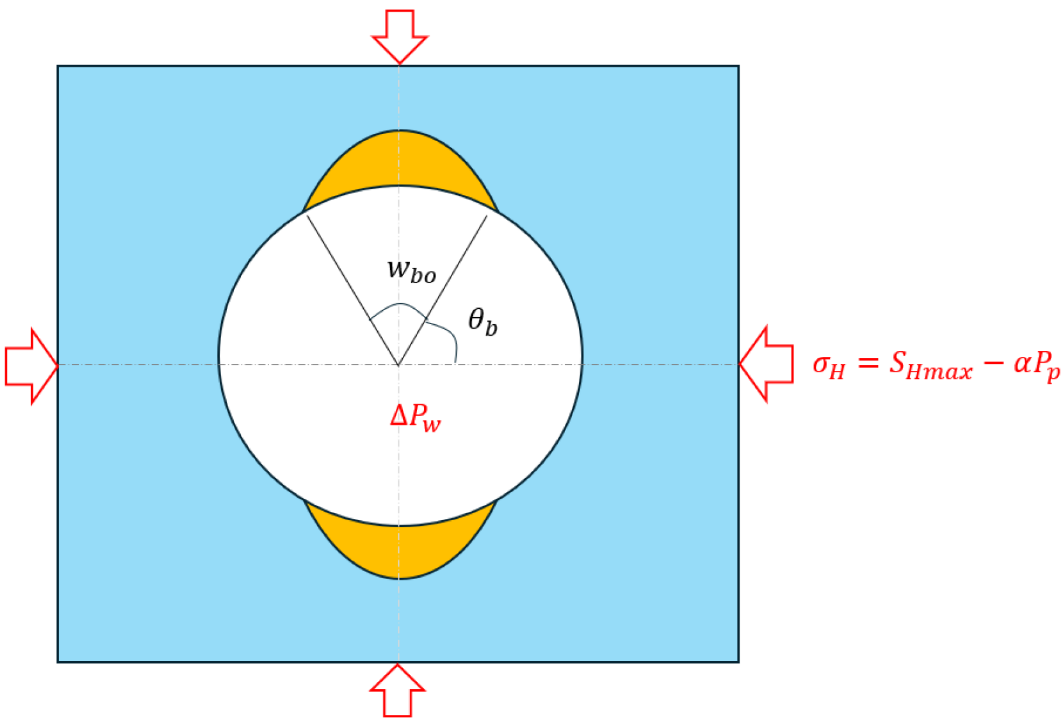

Figure A1—Typical loading condition of impermeable boreholes.

Before any plastic yielding takes place, the radial and tangential stresses on the wellbore surface can be evaluated from Kirsch's solution as:

$$\sigma_r = \Delta P_w \tag{A1}$$

$$\sigma_\theta = \sigma_H + \sigma_h - 2(\sigma_H - \sigma_h)\cos 2\theta - \Delta P_w \tag{A2}$$

Assuming the breakout regions on the wellbore surface run into compressive shear yielding simultaneously, at $\theta = \theta_b$, the effective tangential stress and radial stress shall satisfy Mohr-Coulomb failure criterion:

$$\sigma_\theta = \sigma_r \frac{1+\sin\varphi}{1-\sin\varphi} + UCS \tag{A3}$$

where φ is the friction angle of the formation. Substituting Eq. (A1) and (A2) into Eq. (A3) gives:

$$\sigma_H = \frac{UCS + \frac{2}{1-\sin\varphi}\Delta P_W}{1-2\cos 2\theta_b} - \sigma_h \frac{1+2\cos 2\theta_b}{1-2\cos 2\theta_b} \tag{A4}$$

Assuming Biot's coefficient (α) to be 1 in failure analysis, S_{Hmax} can be obtained by substituting $\sigma_H = S_{Hmax} - P_p$ and $\sigma_h = S_{hmin} - P_p$ into Eq. (A4):

$$S_{Hmax} = \frac{UCS + \frac{2}{1-\sin\varphi}\Delta P_W + 2P_p}{1-2\cos 2\theta_b} - S_{hmin}\frac{1+2\cos 2\theta_b}{1-2\cos 2\theta_b} \tag{A5}$$

For a permeable wellbore, the net wellbore pressure ΔP_w will diffuse into the formation, which will cause stress relaxation in the formation. The loading condition for a permeable borehole in effective stress space is illustrated in Fig. A2. Note, in this case the effective radial stress on wellbore surface becomes zero.

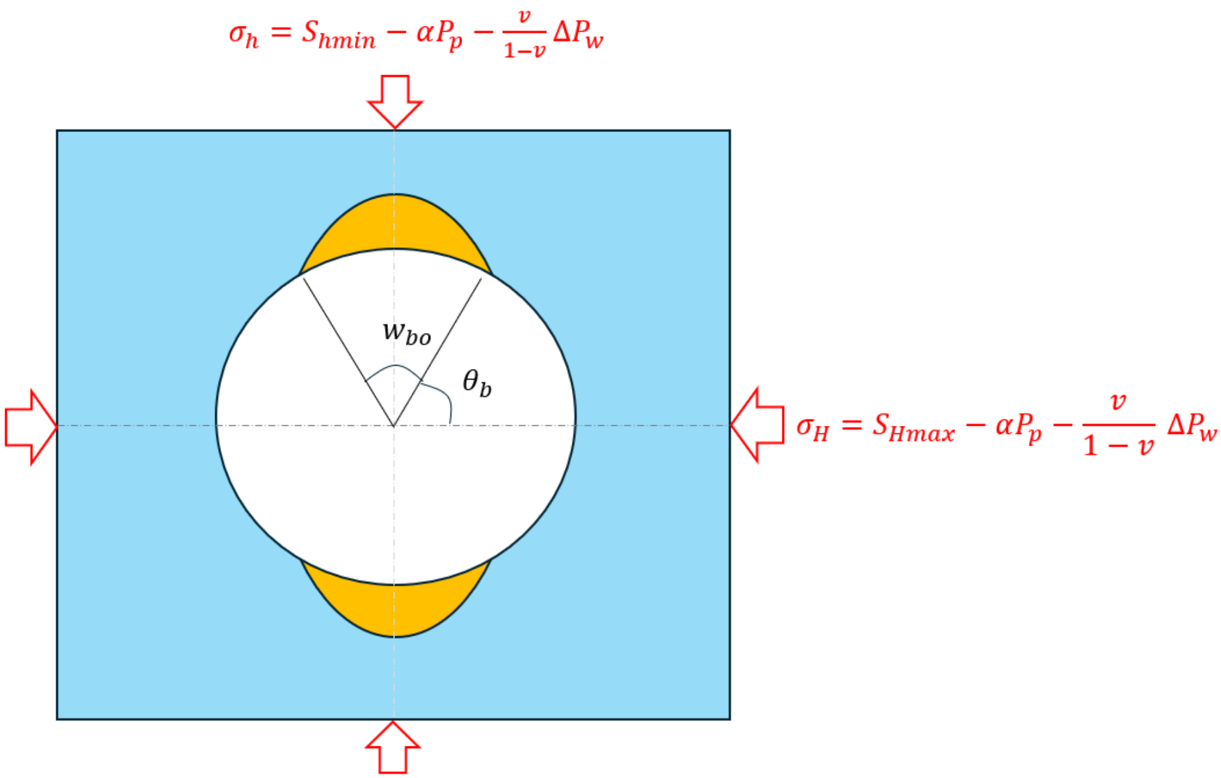

Figure A2—Typical loading condition of permeable boreholes.

After breakout takes place, the effective tangential stress at $\theta = \theta_b$ shall be:

$$\sigma_H = \frac{UCS}{1-2\cos 2\theta_b} - \sigma_h\frac{1+2\cos 2\theta_b}{1-2\cos 2\theta_b} \tag{A6}$$

S_{Hmax}can be solved by substituting $\sigma_H = S_{Hmax} - P_p - \frac{v}{1-v}\Delta P_w$ and $\sigma_h = S_{hmin} - P_p - \frac{v}{1-v}\Delta P_w$ into Eq. (A6) and re-arranging:

$$S_{Hmax} = \frac{UCS + \frac{2v}{1-v}\Delta P_W + 2P_p}{1-2\cos 2\theta_b} - S_{hmin}\frac{1+2\cos 2\theta_b}{1-2\cos 2\theta_b} \tag{A7}$$